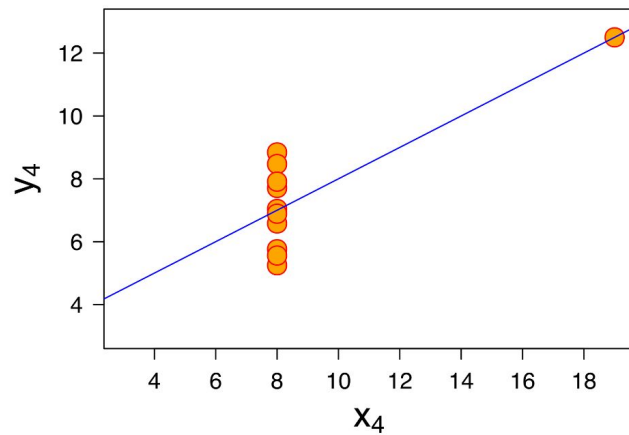
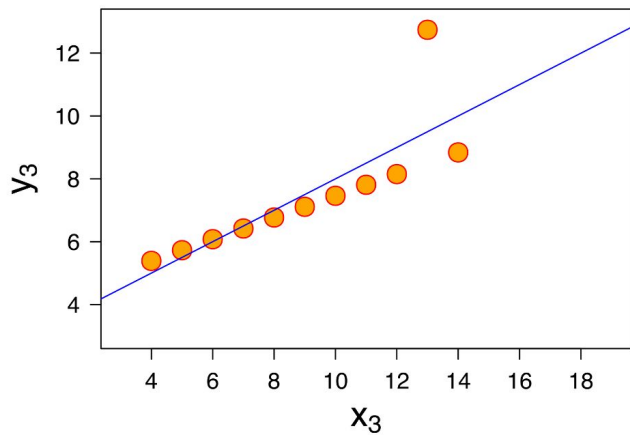
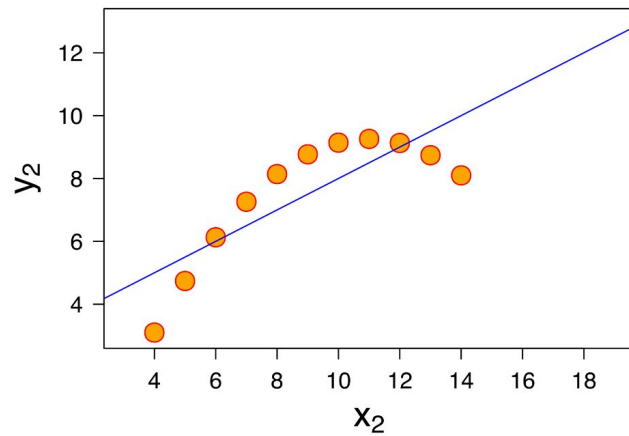
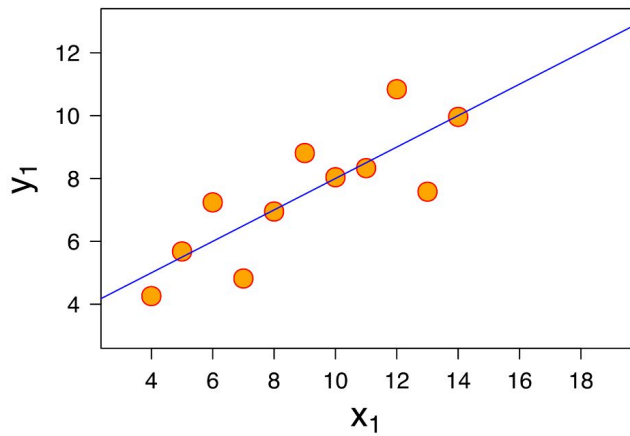


DS-3001

Data Analysis and Visualization

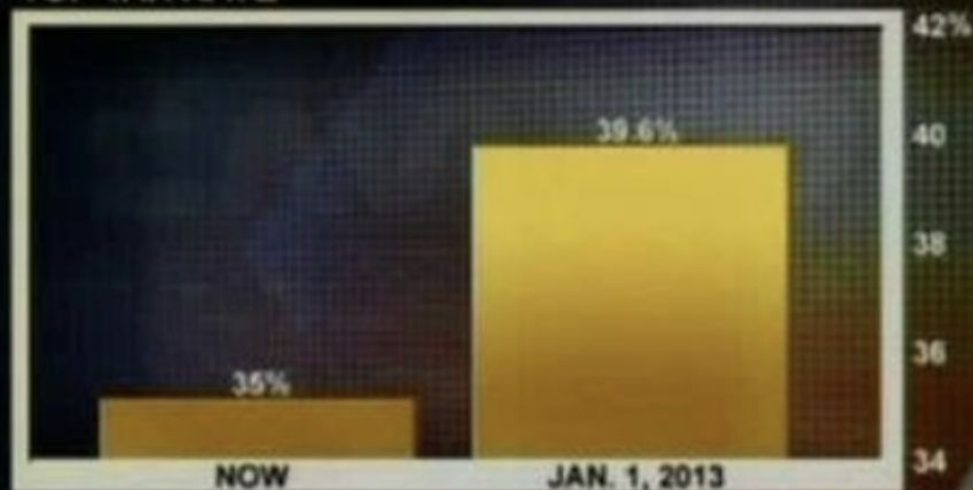
Introduction

	Dataset A	Dataset B	Dataset C	Dataset D
Mean X	9	9	9	9
Variance of X	11	11	11	11
Mean Y	7.50	7.50	7.50	7.50
Variance of Y	4.125	4.125	4.125	4.125
Correlation between X and Y	0.816	0.816	0.816	0.816



IF BUSH TAX CUTS EXPIRE

TOP TAX RATE



8:01 p ET

FOX
BUSINESS

TOP STORIES

TECHNOLOGY

CONSUMER

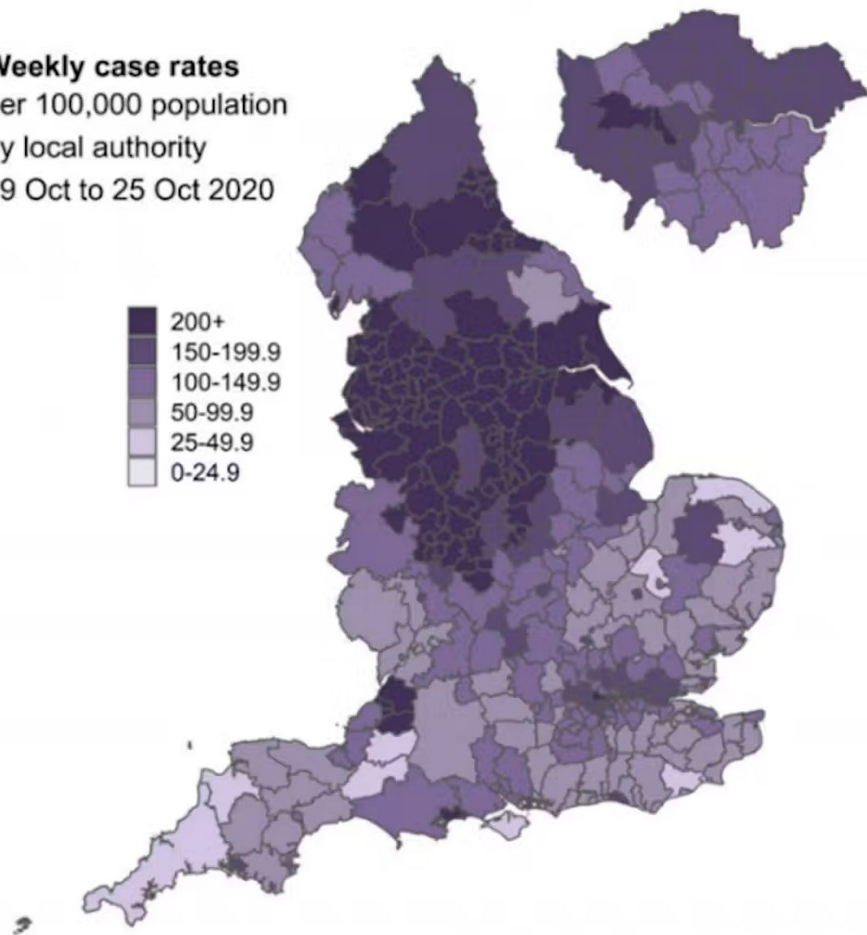
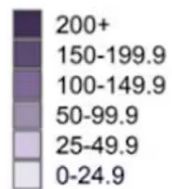
WITH THE JUSTICE DEPARTMENT AND ACQUIRES FULL T

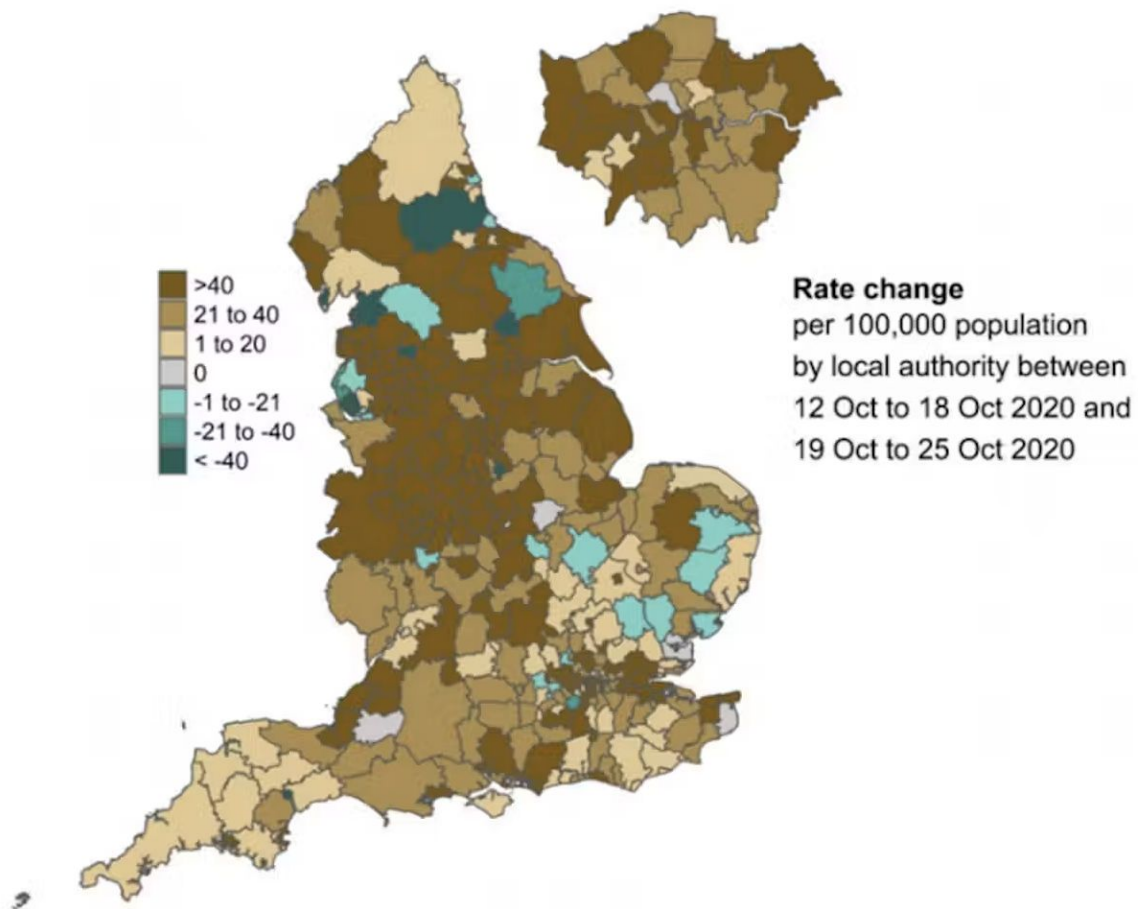
DOW 13008.68 ∇ 64.33

S&P 1379.32 ∇ 5.98

NASDAQ 2939.52 ∇ 6.32

Weekly case rates
per 100,000 population
by local authority
19 Oct to 25 Oct 2020





Kansas COVID-19 7-Day Rolling Average of Daily Cases/Per 100K Population

Mask Counties Vs. No-Mask Mandate Counties

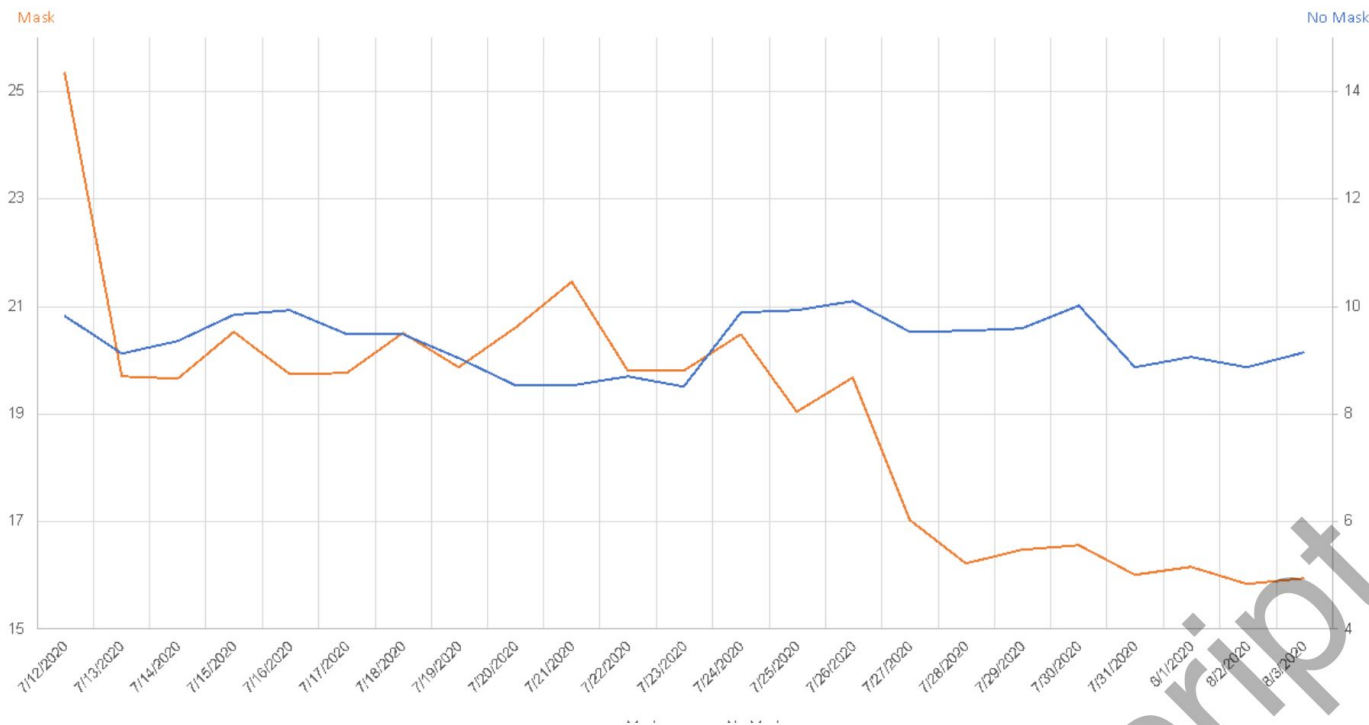
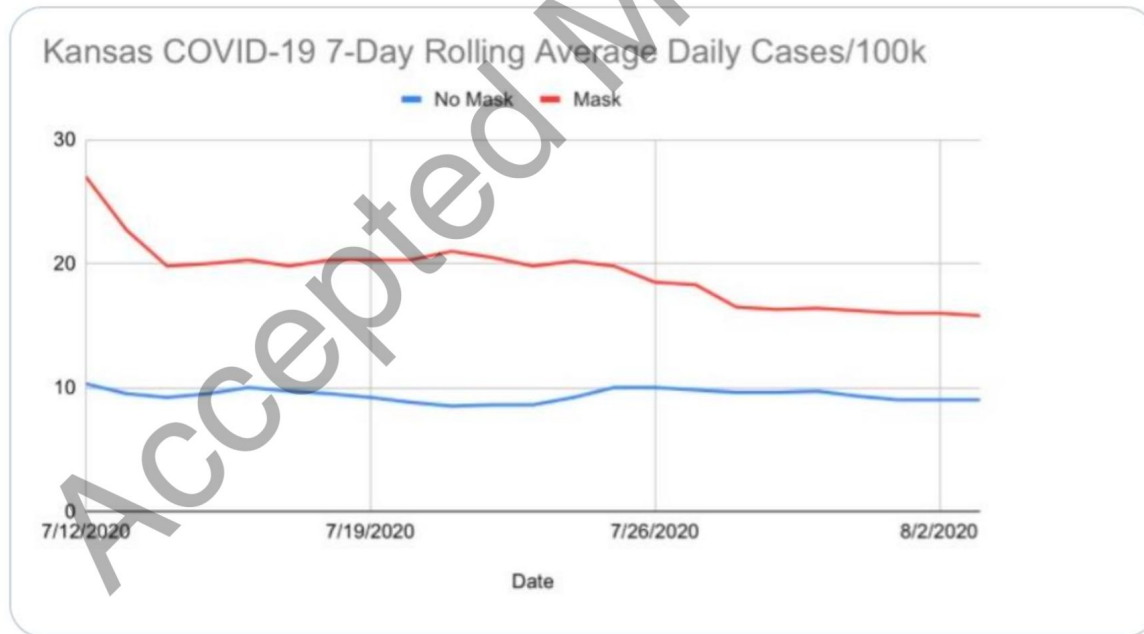


Figure 1. Plot shared by Rachel Maddow on Twitter and live on The Rachel Maddow Show on August 6th, 2020.



Steven Strogatz ✓ @stevenstrogatz · Aug 6

Here's what the graph looks like when plotted the usual way, i.e., on a single vertical scale that includes the origin (courtesy of @quantKid):



9

8

62



Figure 2. Steven Strogatz's Twitter comment to show a recreation of a plot showing the number of daily cases of COVID-19 per 100,000 in the population of Kansas.

Ticket Trend

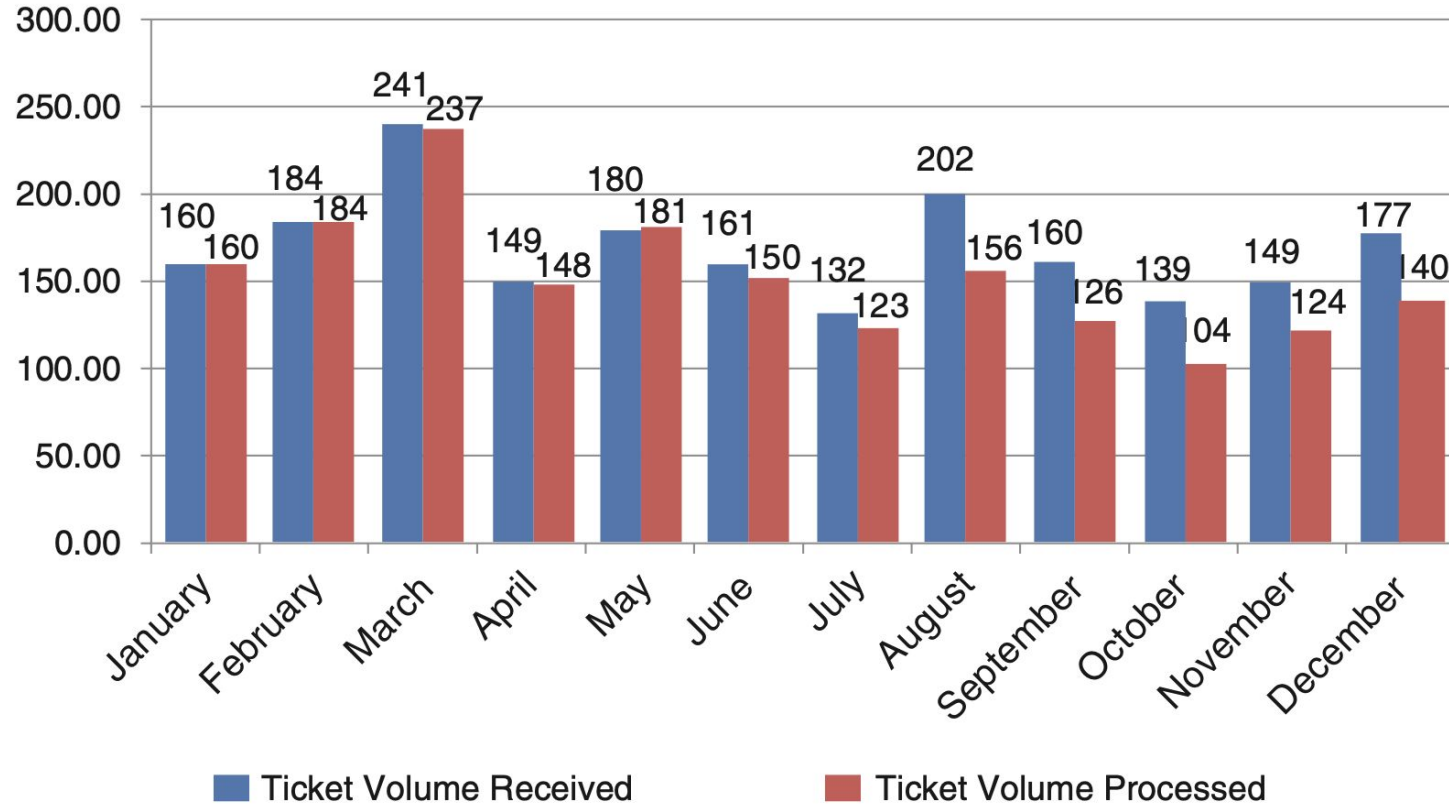


FIGURE 0.2 Example 1 (before): showing data

Please approve the hire of 2 FTEs

to backfill those who quit in the past year

Ticket volume over time



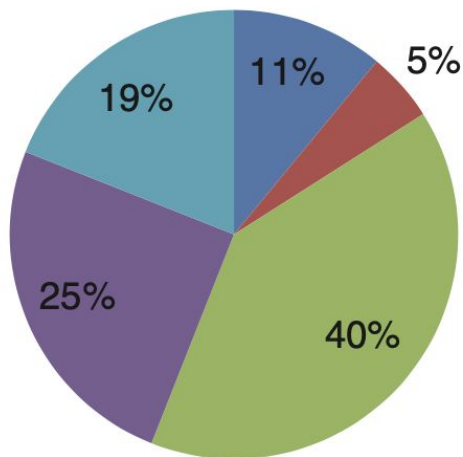
Data source: XYZ Dashboard, as of 12/31/2014 | A detailed analysis on tickets processed per person and time to resolve issues was undertaken to inform this request and can be provided if needed.

FIGURE 0.3 Example 1 (after): storytelling with data

Survey Results

PRE: How do you feel about doing science?

■ Bored ■ Not great ■ OK ■ Kind of interested ■ Excited



POST: How do you feel about doing science?

■ Bored ■ Not great ■ OK ■ Kind of interested ■ Excited

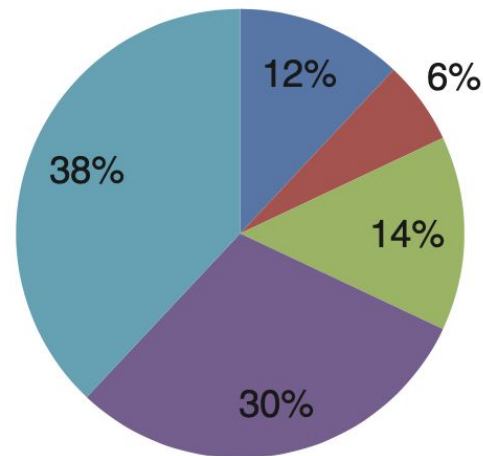
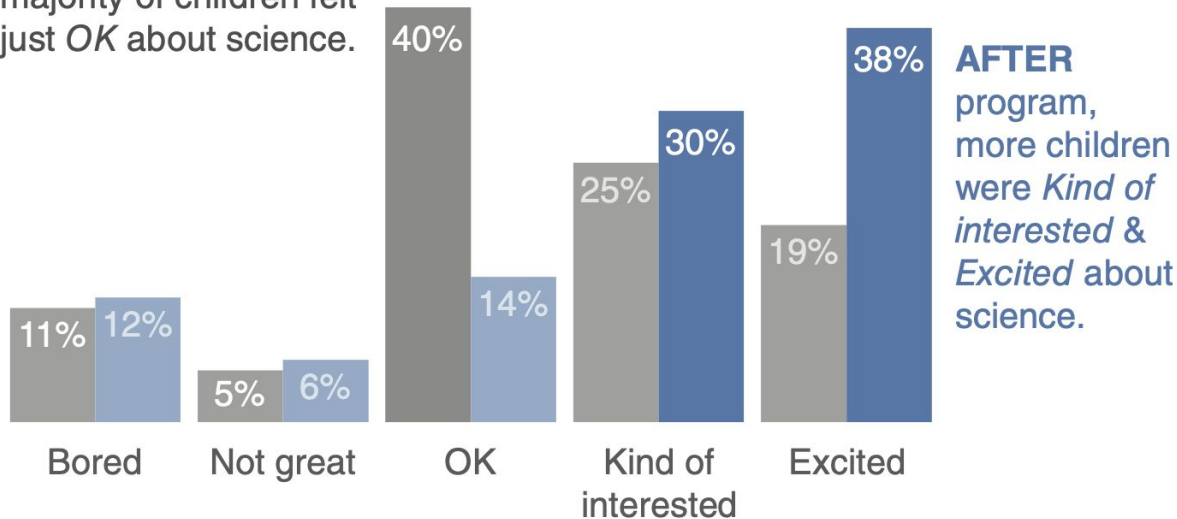


FIGURE 0.4 Example 2 (before): showing data

Pilot program was a success

How do you feel about science?

BEFORE program, the majority of children felt just *OK* about science.



Based on survey of 100 students conducted before and after pilot program (100% response rate on both surveys).

FIGURE 0.5 Example 2 (after): storytelling with data

Average Retail Product Price per Year

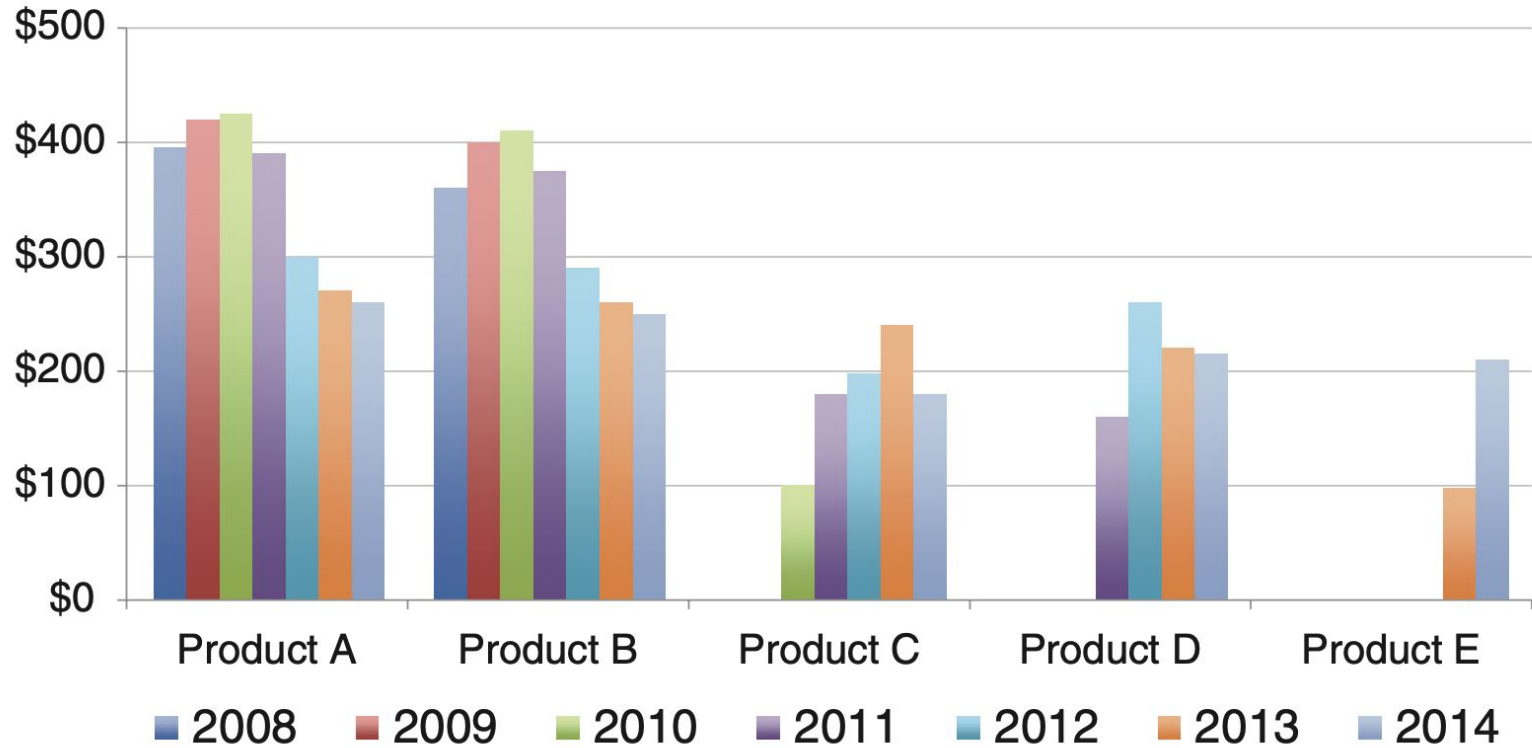


FIGURE 0.6 Example 3 (before): showing data

To be competitive, we recommend introducing our product *below the \$223 average price point* in the **\$150–\$200 range**

Retail price over time by product

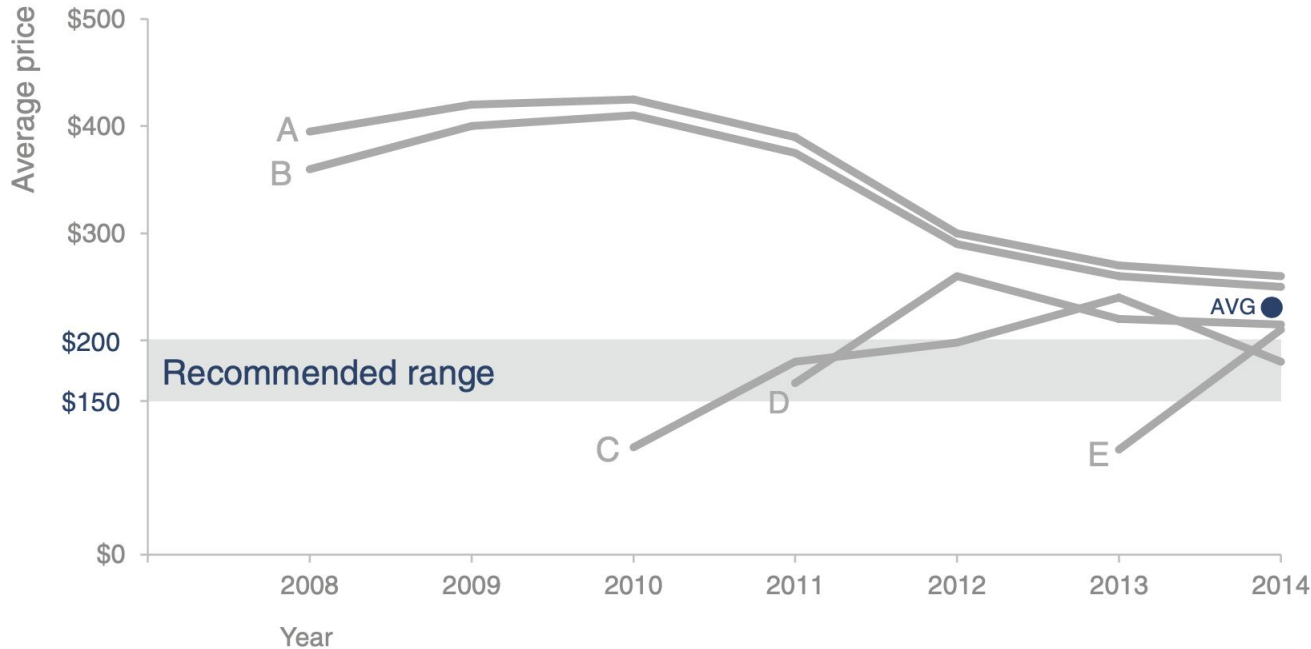


FIGURE 0.7 Example 3 (after): storytelling with data

References

1. https://en.wikipedia.org/wiki/Anscombe's_quartet
2. <https://www.rgs.org/about-us/what-is-geography/geovisualisation/data-visualisation-expert-on-what-s-wrong-with-the-uk-government-s-coronavirus-charts>
3. <https://flowingdata.com/2012/08/06/fox-news-continues-charting-excellence/>
4. https://www.researchgate.net/publication/350825789_Data_Misrepresentation_and_COVID-19_Leveraging_Misleading_Data_Visualizations_For_Developing_Statistical_Literacy_Across_Grades_6-16
5. Storytelling with Data- A Data Visualization Guide for Business Professionals by Cole Nussbaumer Knaflic